

SDN CONTROLLER EVALUATION FOR CAMPUS DEPLOYMENTS

Comparative Analysis: OpenDaylight (ODL) vs. Floodlight

DIRECT RECOMMENDATION

OpenDaylight (ODL) is the significantly better choice for a campus network requiring an enterprise-grade solution with a comprehensive feature set and extensive network service support. While Floodlight is lightweight and well-suited for academic experimentation or smaller-scale setups, OpenDaylight is architected specifically for complex, large-scale multi-protocol enterprise environments.

KEY ADVANTAGES OF OPENDAYLIGHT (ODL) FOR CAMPUS NETWORKS

- **Multi-Protocol Southbound Support:** Unlike Floodlight, which is strictly focused on OpenFlow, OpenDaylight natively supports a wide range of southbound protocols including OpenFlow, NETCONF, OVSDB, BGP-LS, PCEP, and SNMP. This allows the university to integrate both modern SDN switches and existing legacy campus routing/switching hardware.
- **Model-Driven Service Abstraction Layer (MD-SAL):** ODL utilizes YANG data modeling and MD-SAL to auto-generate APIs and logically separate network applications from underlying hardware. This provides flexible orchestration across heterogeneous campus network services.
- **Enterprise-Grade Scalability & High Availability:** ODL supports multi-node clustering, distributed datastores, and robust failover mechanisms, ensuring high availability and fault tolerance necessary for mission-critical university operations and campus-wide deployments.
- **Comprehensive Suite of Built-in Services:** ODL provides an extensive ecosystem of enterprise network services, including automated AAA (Authentication, Authorization, and Accounting), L2/L3 VPN orchestration, microsegmentation security policies, dynamic traffic engineering, and seamless cloud integration (e.g., OpenStack Neutron).
- **Strong Industry Backing & Ecosystem:** Supported by major networking industry leaders (such as Cisco, VMware, and Red Hat) under the Linux Foundation, ODL offers a mature, production-ready framework with long-term maintenance and community support.

FLOODLIGHT LIMITATIONS IN ENTERPRISE SCENARIOS

Floodlight is primarily an OpenFlow-centric Java controller designed for simplicity and light footprint. It lacks built-in multi-protocol support, clustering capability, advanced network abstraction models, and out-of-the-box enterprise-grade management features needed for a heterogeneous university campus.

CONCLUSION

For an enterprise-grade campus deployment requiring diverse network services, seamless hardware integration, and high availability, **OpenDaylight (ODL)** stands out as the optimal SDN controller.